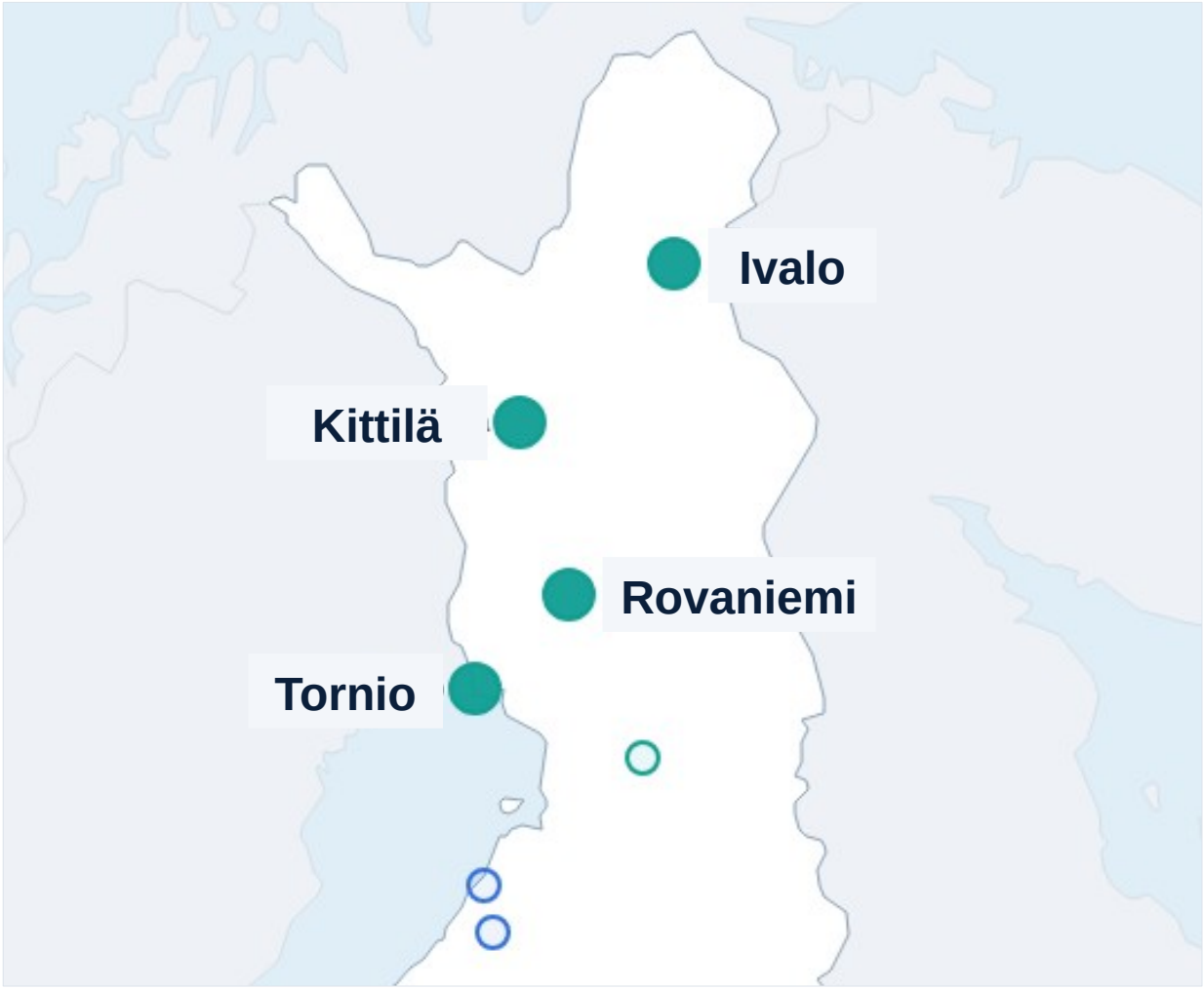
Tulva

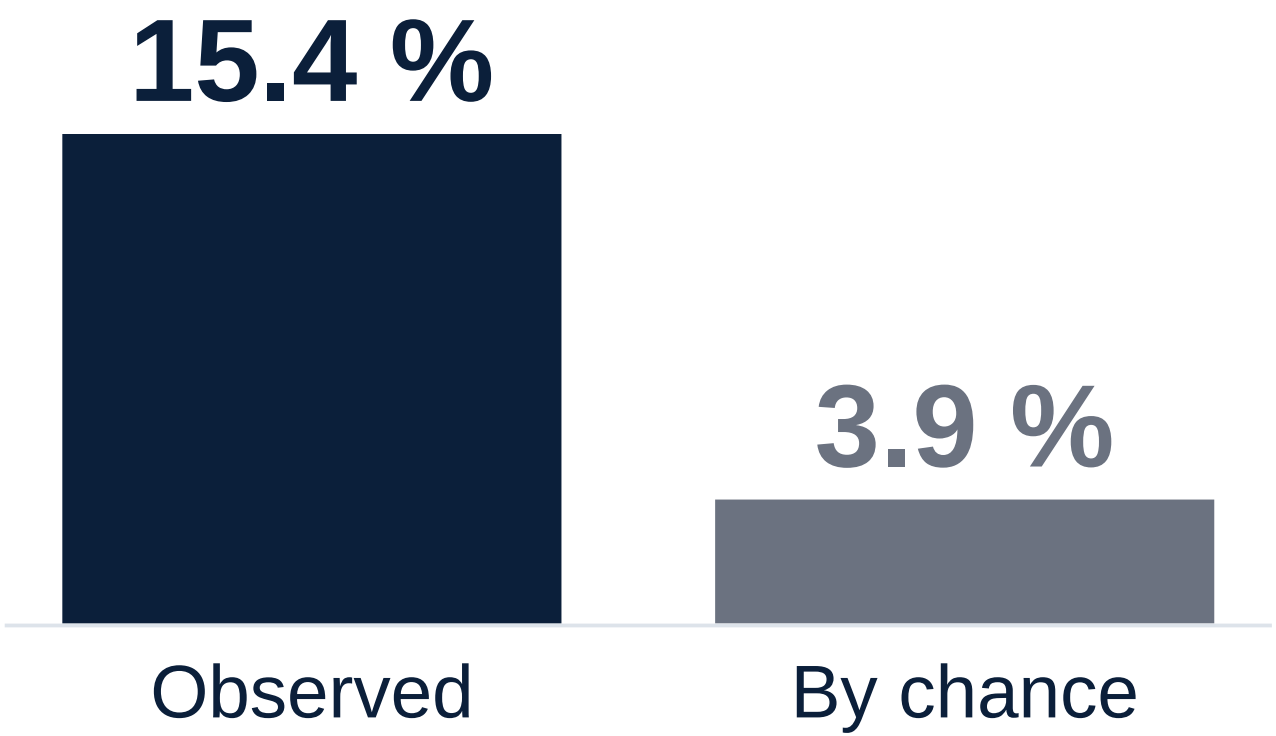
May 2005: four Lapland rivers flooded together.

The March snow warned us.



Rivers flood together, not one at a time

Springs when ≥4 of 14 rivers flood

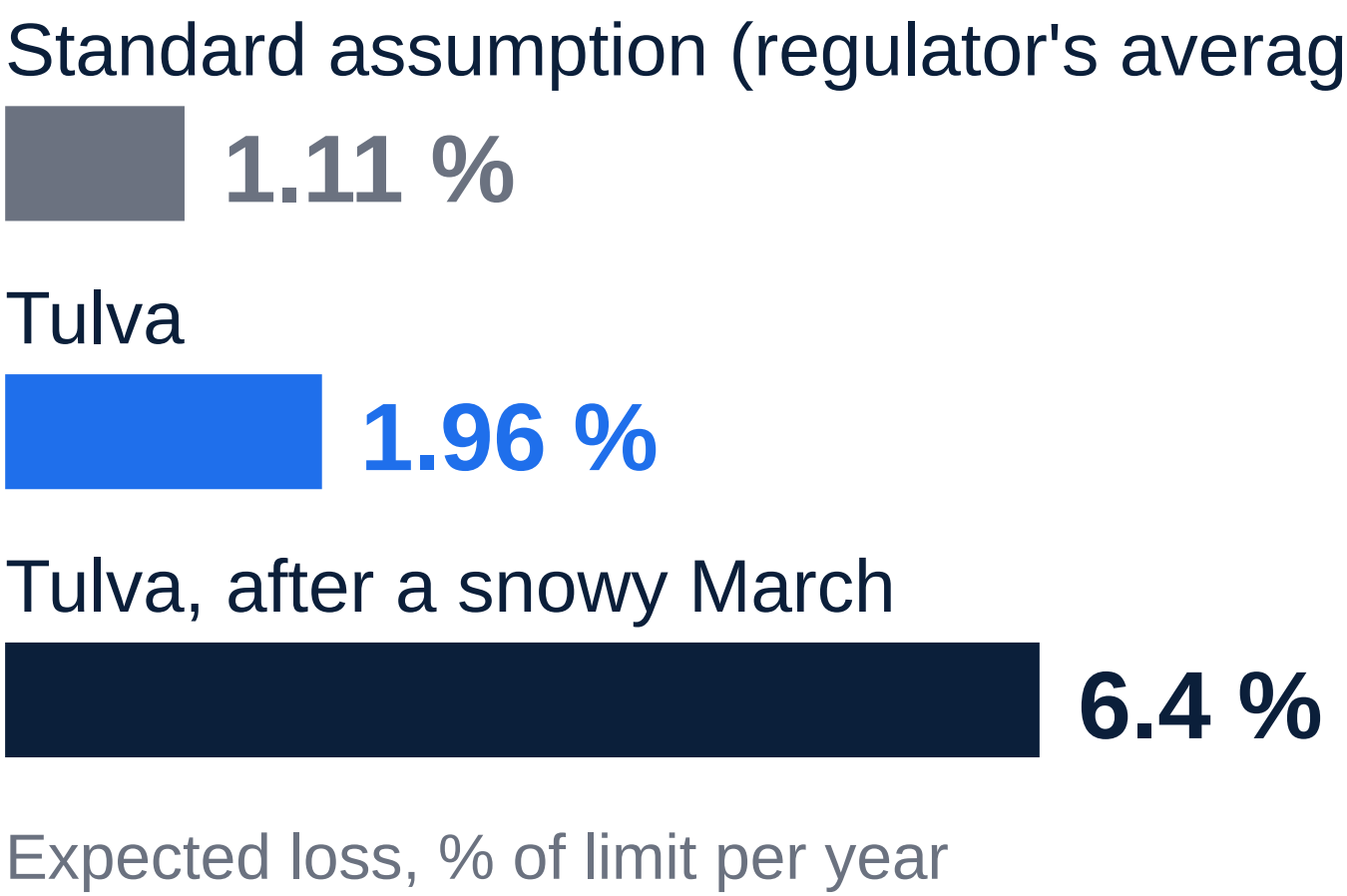


≈4×

as often as chance

Cover is bought in autumn, before the snow.

Tulva Index: pays if ≥2 northern rivers flood



×3

expected loss after a snowy March

FINLAND · 14 RIVER FLOOD-RISK AREAS

May 2005

4 of 14 areas

over their 1-in-10 flood level, together

Two months earlier: Lapland's snow on 1 March

top 15 %

least snow — most snow (1961–2025)

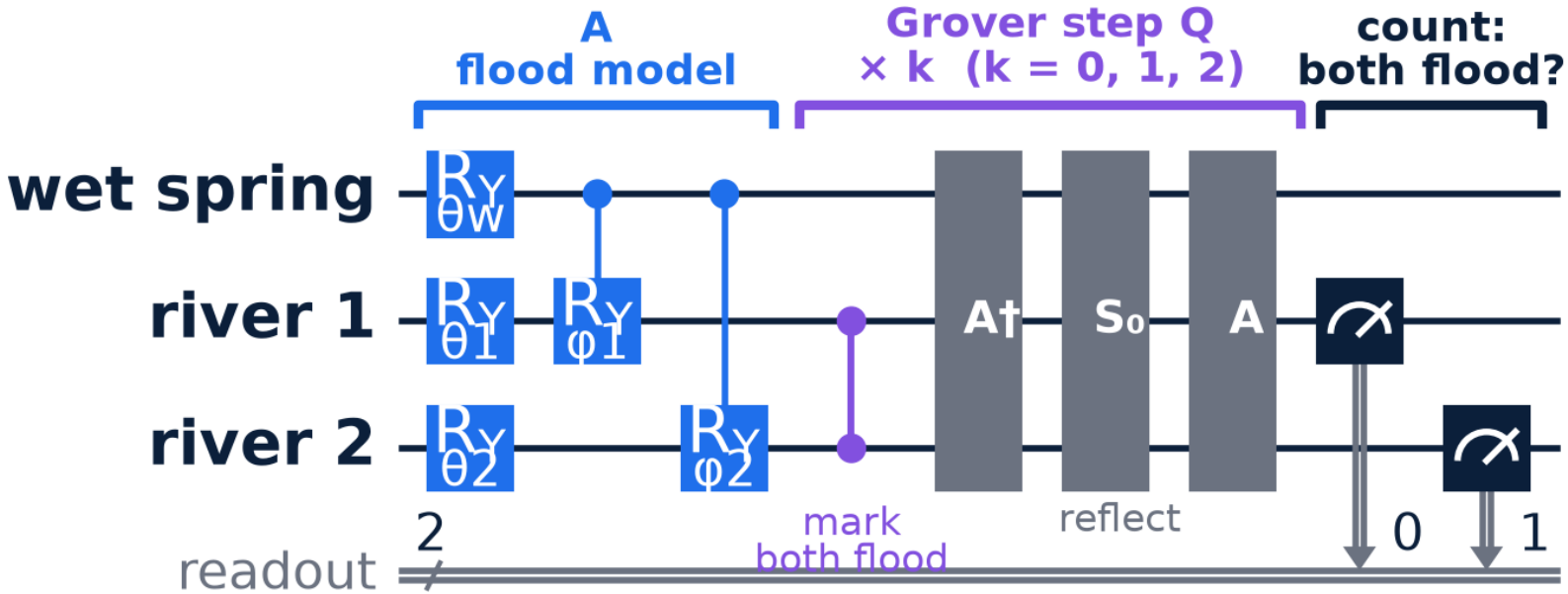
87th percentile · top 15 %



SYKE river and snow data · peak spring flow vs each gauge's 1-in-10 level · back-test, not a forecast made in 2005

the March snow was in its top 15 %.

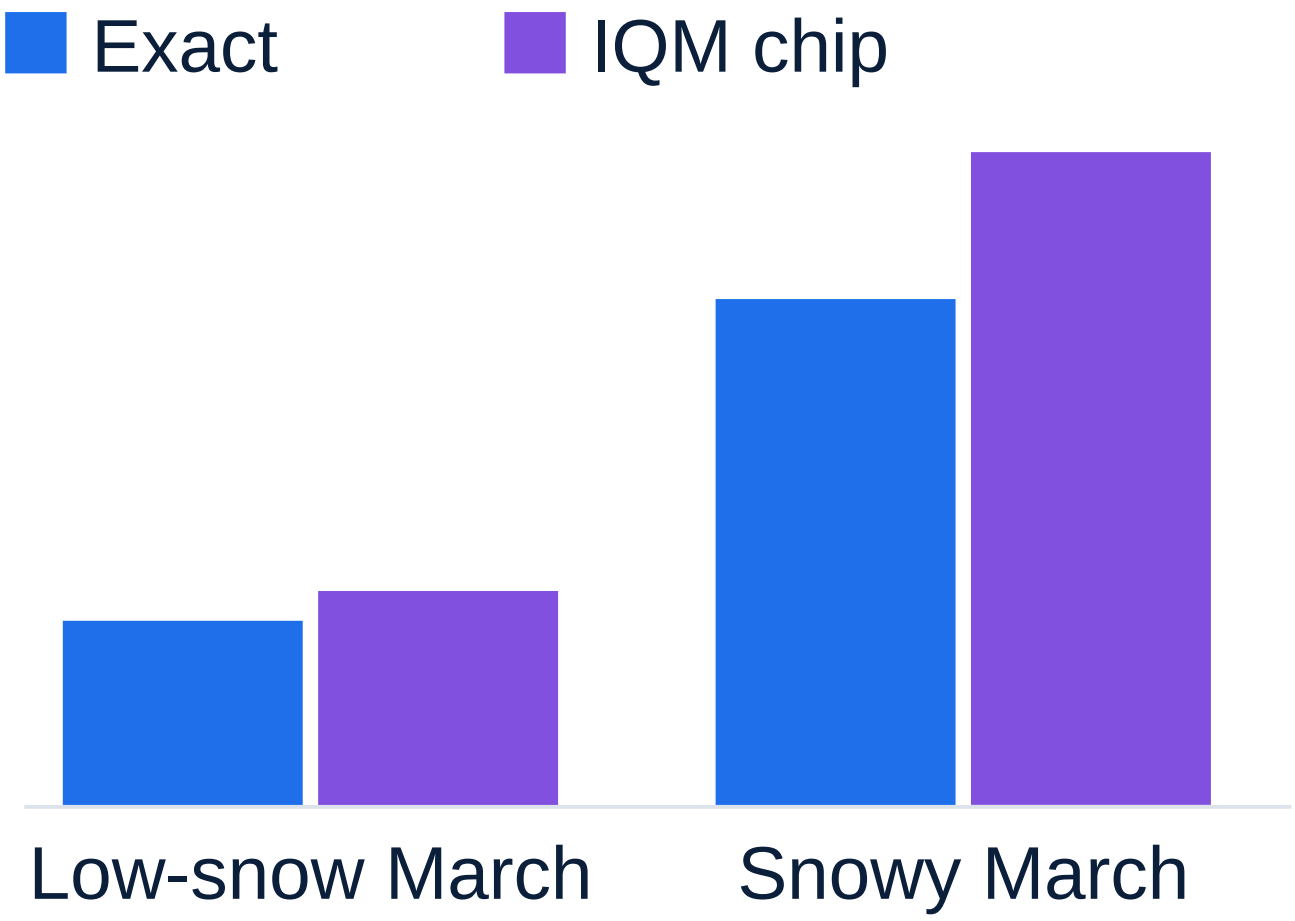
Our circuit: quantum amplitude estimation



Qubits = wet spring + rivers. Gates = our flood model.

Why quantum? 2× accuracy: sampling needs 4× the runs, QAE 2×.

On real chips, the snow effect still shows



Snowy vs low-snow March

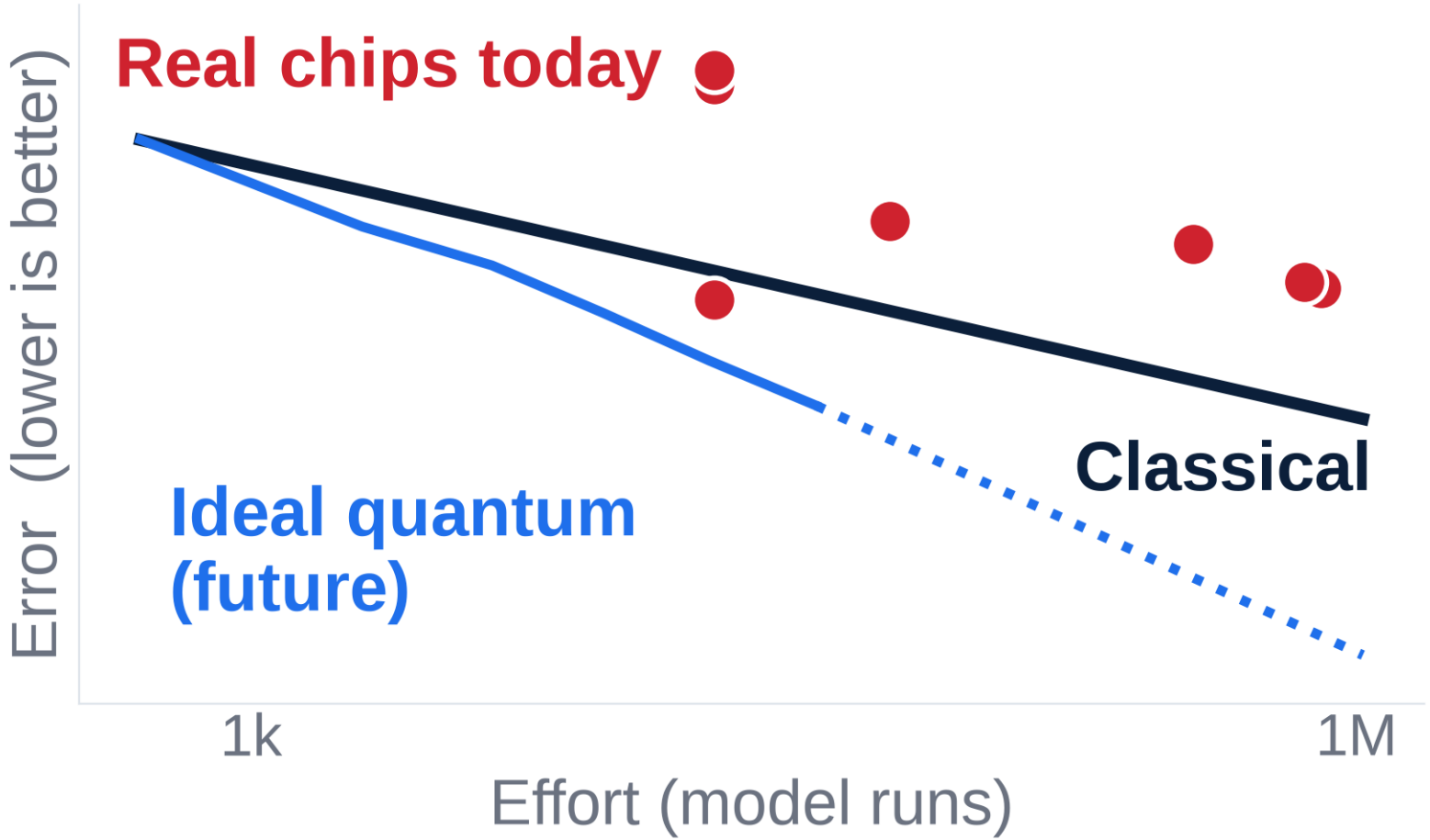
3.05× on chip

2.75× exact

Ran on

IBM · IQM · VTT Q50

Classical still wins today



Rare 1-in-50 floods drown in noise.

Prices are computed classically.

Who pays



€25k pilot, then €75k / yr

Year 3: €0.2–0.45M / yr

3–6 Finnish insurers, then the Nordics

Ask: one insurer's flood claims, 8 weeks.

Built this weekend

SYKE data pipeline

Joint flood model

Index pricing

Live app

Runs on 4 quantum chips

Next March, the snow will warn us again.

This time, it will have a price.

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026